When Sample Selection Bias Precipitates Model Collapse

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Introduction

Recursive synthetic-data loops have two canonical baselines: the *Replace paradigm* (Shumailov et al., Nature 2024), which discards historical data and can eliminate variance, and the *Accumulate paradigm* (Kazdan et al., ICML 2025), which appends generations and is stable without biased filtering.

Theory

Proposition 1: Replace paradigm (Shumailov et al., Nature 2024)

Under the *Replace* paradigm, as $t \to \infty$,

$$\boldsymbol{\Sigma}_t \xrightarrow{a.s.} \mathbf{0}, \qquad \mathbb{E}\left[\mathbb{W}_2^2(\mathcal{N}_t, \mathcal{N}_0)\right] \to \infty.$$

Replacing data eventually removes variance and drifts away.

Proposition 2: Accumulate paradigm (Kazdan et al., ICML 2025)

Under unbiased accumulation,

$$\mathbb{E}[\bar{\boldsymbol{\Sigma}}_t] \to \alpha_n \bar{\boldsymbol{\Sigma}}_0.$$

Thus $\mathbb{E}[\mathbb{W}_2^2] \to 2(1-\sqrt{\alpha_n})\,\mathrm{Tr}(\bar{\boldsymbol{\Sigma}}_0)$.

Unbiased accumulation keeps diversity from fully collapsing.

Theorem 1: local selection collapses diversity

With top-α selection toward a local ideal $\mathbf{u}^*$,

$$\|\bar{\boldsymbol{\mu}}_t - \mathbf{u}^*\|^2 \to 0, \qquad \bar{\boldsymbol{\Sigma}}_t \to \mathbf{0}.$$

Local fidelity erases global coverage.

Local selection gains fit by erasing global coverage.

Theorem 2: diversity decays by power law

If the trajectory remains in the local basin,

$$\mathrm{Tr}(\bar{\boldsymbol{\Sigma}}_t) = \mathcal{O}_{a.s.}(t^{-\psi}).$$

Tail erosion has an explicit rate.

Once trapped locally, diversity shrinks at a predictable rate.

Theorem 3: drift controls true-manifold risk

For filtered distribution $\mathcal{D}_t$ and true manifold $\mathcal{D}^*$,

$$\mathcal{R}_{\mathcal{D}^*}(h_t; g^*) \le 2\ell\epsilon\,\mathbb{W}_p(\mathcal{D}_t, \mathcal{D}^*) + \mathcal{R}_{\mathcal{D}_t}(h_t; g_t) + \mathcal{O}(\ell\delta).$$

Distribution drift upper-bounds risk on true data.

For more methodology and theoretical details, please refer to the full paper.

Biased selection makes AI models forget rare cases.

Scan

OpenReview paper and supplement.

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Multivariate Gaussian Modeling

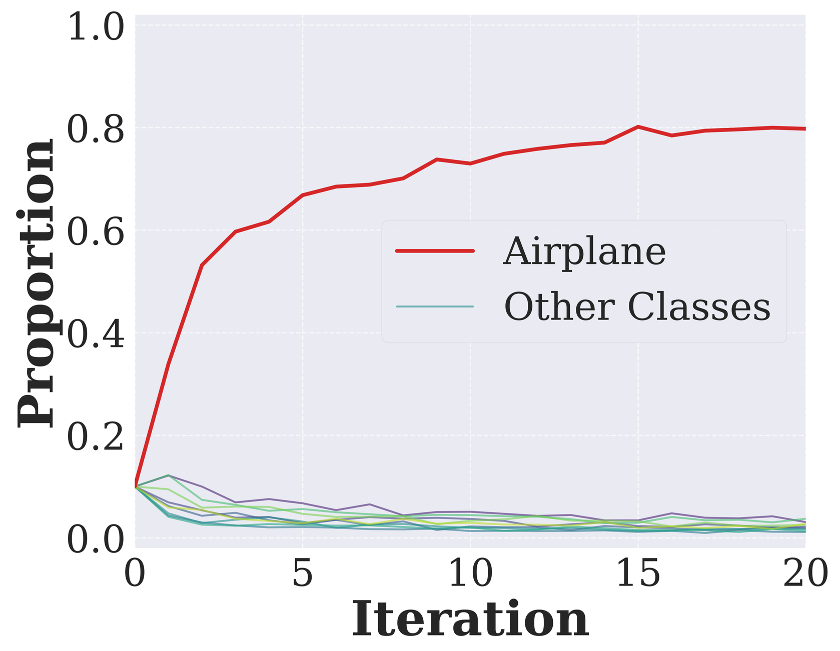
Gaussian (main text). Biased top-α selection collapses variance in both recursive baselines; accumulation slows collapse but follows the predicted power-law decay.

Anisotropic Gaussian. Direction-dependent spread still contracts when the verifier favors a restricted local region.

Gaussian mixture. Multimodal and imbalanced mixtures lose component coverage, with minority modes eroding first.

Laplace. Heavy-tailed scale contracts sharply, showing tail loss beyond Gaussian variance.

Collapse Signal



Airplane-only verification drives CIFAR-10 proportions toward Airplane, exposing systematic tail-mode pruning.

Experimental Synthesis

Across CIFAR-10, STL-10, and CelebA, skewed local references can make selection weaker than random sampling; broader distributional evidence improves FID, precision, and recall.

References

[1] Shumailov et al. AI models collapse when trained on recursively generated data. Nature, 2024.

[2] Kazdan et al. Collapse or Thrive: Perils and Promises of Synthetic Data in a Self-Generating World. ICML, 2025.

For additional experimental results, please refer to the full paper.

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